Markov decision processes

Model uncertain outcomes. Compute a policy.

Understand the model

States · actions · outcomes
Rewards · policies · return

Solve the model

Bellman equations
Value and policy iteration

35 min math & examples · 5 min demo · 5 min recap

From plans to policies

What should we do after an uncertain outcome?

Week 5 · Plan

Choose an action sequence.

→

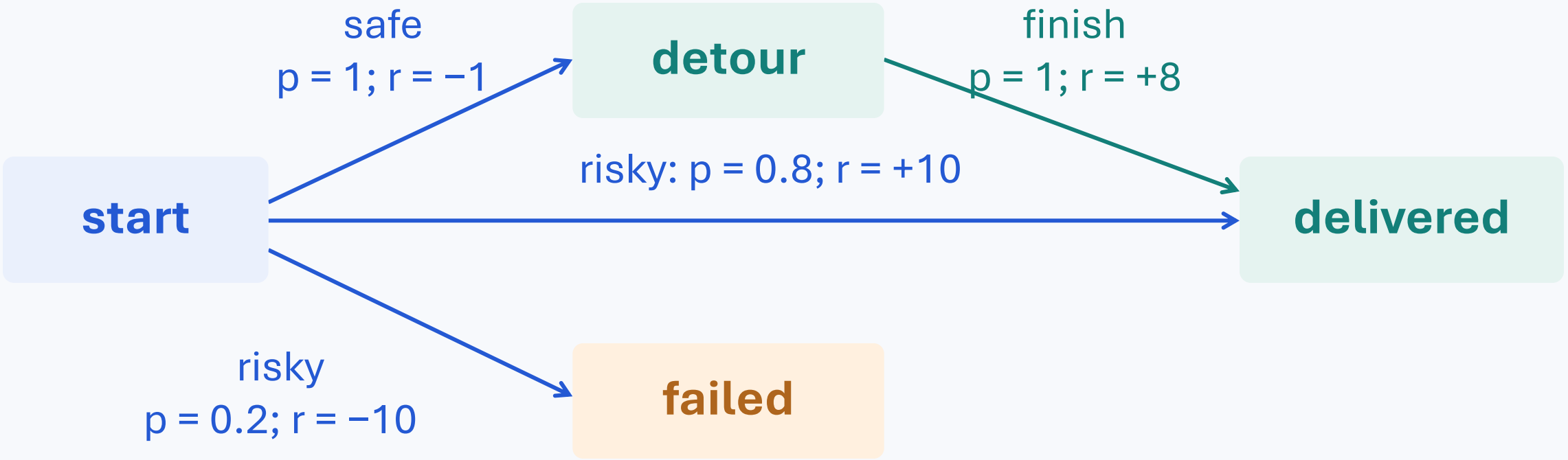
Week 6 · Policy

Choose an action for each observed state.

The robot chooses an action; the world determines its outcome.

Our delivery robot

Two routes. Different outcomes and rewards.



γ = 0.9 · Rewards belong to transitions.

Define every part of the MDP

A Markov decision process is a mathematical model.

$$\mathcal{M} = (S, \{A(s)\}, P, r, \gamma)$$

S · States	start, detour, delivered, failed
A(s) · Actions	safe / risky; finish; stay
P · Outcomes	Next-state probabilities
r · Rewards	Reward on each transition
γ · Discount	Future-reward weight: 0.9

s: one state | S_t: random state at time t

Read transition probabilities

Condition on the current state and the chosen action.

$$P(s' \mid s, a) \geq 0 \quad \sum_{s' \in S} P(s' \mid s, a) = 1$$

$$P(\text{delivered} \mid \text{start}, \text{risky}) = 0.8$$

$$P(\text{failed} \mid \text{start}, \text{risky}) = 0.2$$

The two risky outcomes sum to 1. Other successors have probability 0.

What makes a state Markov?

The present state must contain the relevant information.

$$P(S_{t+1} \mid S_{0:t}, A_{0:t})$$

$$= P(S_{t+1} \mid S_t, A_t)$$

If battery changes success probability, include battery in the state.

A policy chooses actions

A deterministic decision can have a stochastic outcome.

Deterministic: $\pi(s) \in A(s)$

Stochastic: $\pi(a \mid s) \geq 0, \ \sum_a \pi(a \mid s) = 1$

Example: $\pi(\text{start}) = \text{risky}$

At detour: finish. At either terminal: stay.

Reward now; value of the future

Do not count the arrival reward twice.

$r(s,a,s')$: reward on this transition

$$R(s,a) = \sum_{s'} P(s' \mid s,a)\, r(s,a,s')$$

$$R(\text{start},\text{risky}) = 0.8(10) + 0.2(-10) = 6$$

Delivered / failed: later rewards = 0, so terminal values = 0.

Return and discounting

Return adds rewards along one trajectory.

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k}$$

$$G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \cdots$$

$$\text{Safe:} \quad G = -1 + 0.9(8) = 6.2$$

Risky: one trajectory returns +10 or −10. Expected return is 6.

State values and action values

Both average future return under a policy.

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$$

$$Q^{\pi}(s,a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a]$$

V: follow π now. Q: take a now, then follow π.

Derive the Bellman recursion

One reward now, then the same problem at the next state.

$$G_t = R_t + \gamma G_{t+1}$$

$$V^\pi(s) = \mathbb{E}_\pi[R_t + \gamma G_{t+1} \mid S_t = s]$$

$$= \mathbb{E}_\pi[r(s, A_t, S_{t+1}) + \gamma V^\pi(S_{t+1}) \mid S_t = s]$$

Condition on the chosen action and successor; apply the Markov property.

Evaluate a fixed policy

Average actions and outcomes; keep the policy fixed.

$$V^{\pi}(s) = \sum_{a \in A(s)} \pi(a \mid s) \{$$

$$\sum_{s' \in S} P(s' \mid s,a) \, [r(s,a,s') + \gamma V^{\pi}(s')] \}$$

Deterministic: $V^{\pi}(s) = Q^{\pi}(s, \pi(s))$

There is no max: evaluation asks how good this policy is.

Evaluate the risky policy by hand

π(start) = risky; π(detour) = finish.

$$V^{\pi}(\text{delivered}) = V^{\pi}(\text{failed}) = 0$$

$$V^{\pi}(\text{detour}) = 1[8 + 0.9(0)] = 8$$

$$V^{\pi}(\text{start}) = 0.8[10+0.9(0)] + 0.2[-10+0.9(0)]$$

$$= 6$$

The policy's value at detour is defined even if start → risky never visits it.

Policy evaluation as linear algebra

One equation per nonterminal state.

$$P_\pi(i,j) = \sum_a \pi(a \mid i)\, P(j \mid i,a)$$

$$R_\pi(i) = \sum_a \pi(a \mid i) \sum_{s'} P(s' \mid i,a)\, r(i,a,s')$$

$$V^\pi = R_\pi + \gamma P_\pi V^\pi$$

$$(I - \gamma P_\pi) V^\pi = R_\pi$$

Vector order: [start, detour]. Solve a 2 × 2 linear system.

Solve the safe policy's system

Rows and columns use [start, detour].

$$P_\pi = [0 \;\; 1; 0 \;\; 0] \qquad R_\pi = [-1 \;\; 8]^T$$

$$\begin{bmatrix} 1 & -0.9 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V^\pi(\text{start}) \\ V^\pi(\text{detour}) \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \end{bmatrix}$$

V(detour) = 8 ⇒ V(start) = −1 + 0.9(8) = 6.2

Choose actions; average outcomes

Bellman optimality separates control from uncertainty.

$$Q_V(s,a) = \sum_{s'} P(s' \mid s,a)[r(s,a,s') + \gamma V(s')]$$

$$V^*(s) = \max_{a \in A(s)} Q^*(s,a)$$

$$\text{At start:} \quad \max\{-1 + 0.9(8), 6\} = 6.2$$

Sum over successor states. Max over actions.

Value iteration

Improve values through repeated maximizing backups.

Initialize: $V_0(s) = 0$

$$V_{k+1}(s) = \max_a \sum_{s'} P(s' \mid s,a)[r(s,a,s') + \gamma V_k(s')]$$

$$\Delta_k = \max_s |V_{k+1}(s) - V_k(s)|$$

Read the complete old vector; write a separate new vector.

Value iteration by hand

Every row uses the complete preceding row.

Sweep	start	detour	delivered	failed
V_0	0	0	0	0
V_1	6	8	0	0
V_2	6.2	8	0	0
V_3	6.2	8	0	0

V_1(start): max{−1, 6} = 6. V_2(start): max{6.2, 6} = 6.2.

Extract a policy from the displayed V

The greedy action uses a fresh one-step lookahead.

$$\pi_V(s) \in \operatorname{argmax}_a Q_V(s,a)$$

$$\text{Using } V_1: \quad Q_{V_1}(\text{start},\text{safe}) = -1 + 0.9(8) = 6.2$$

$$\text{Using } V_1: \quad Q_{V_1}(\text{start},\text{risky}) = 6$$

V_1(start) = 6, but the greedy action from V_1 is already safe.

Policy iteration

Evaluate a policy completely, then improve it.

Evaluate: $(I - \gamma P_{\pi_k})V^{\pi_k} = R_{\pi_k}$

Improve: $\pi_{k+1}(s) \in \operatorname{argmax}_a Q^{\pi_k}(s,a)$

Repeat until the policy is unchanged and the optimality residual is small.

Policy iteration by hand

The visualization deliberately starts with risky.

Round 0 · risky

Evaluate: V(start) = 6
Compare: safe = 6.2
Improve: choose safe

Round 1 · safe

Evaluate: V(start) = 6.2
Improve: still safe
Stable policy; residual = 0

Evaluate → improve → evaluate again → unchanged.

Keep three operations separate

Identify what is fixed before reading an equation.

Policy evaluation

Fixed: policy + model
Compute: V^{π}

Value iteration

Fixed: model
Maximizing value backups

Policy iteration

Fixed: model
Evaluate + improve policy

Known model → planning. No probability learning in this demo.

Change risk; change the policy

Predict before selecting failure probability 0.1.

Risky at p=0.2: $0.8(10) + 0.2(-10) = 6$

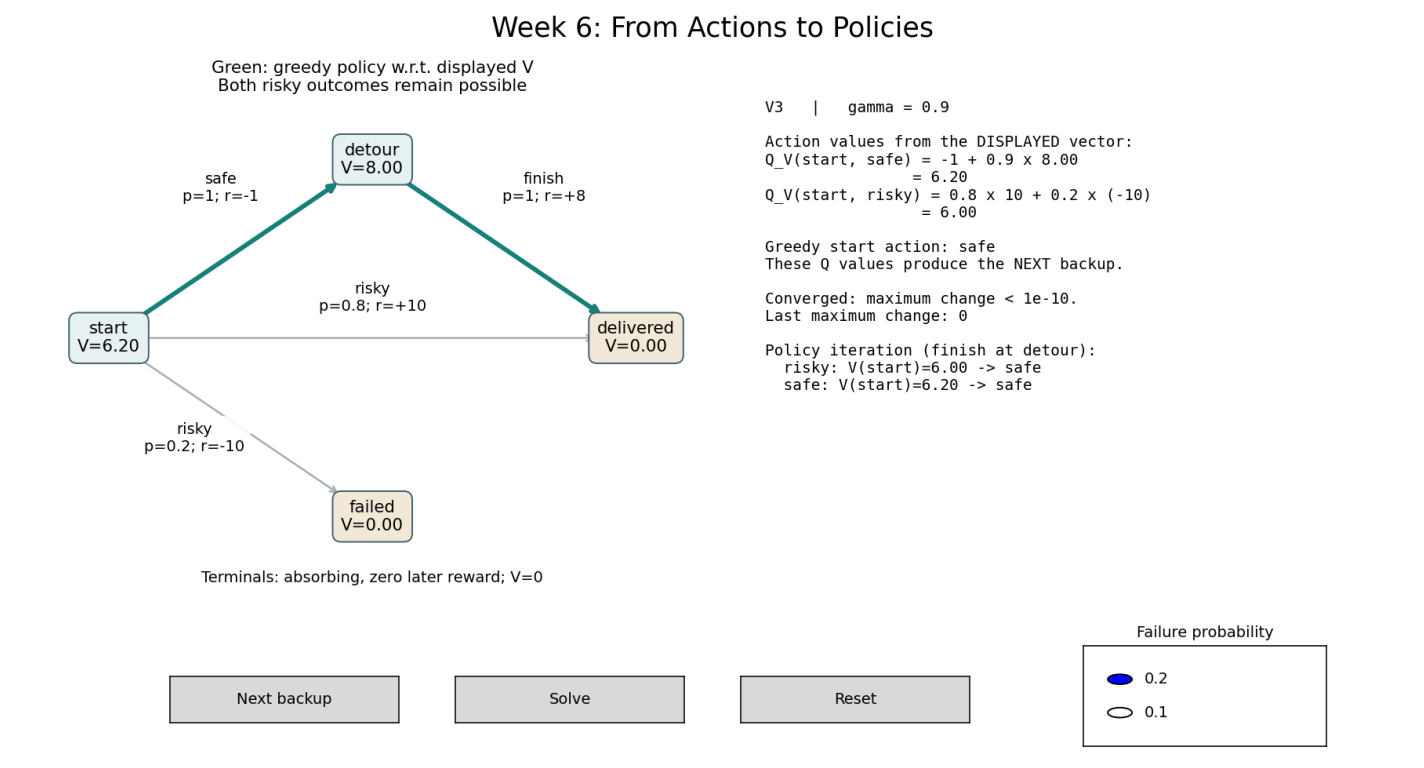
Risky at p=0.1: $0.9(10) + 0.1(-10) = 8$

Safe remains: $-1 + 0.9(8) = 6.2$

The optimal start action changes: safe → risky.

Predict, then visualize

The program applies the equations we just calculated.



Predict the first backup.

Explain the green arrows.

Change risk. Predict again.

Checkpoint & HW3

Explain the calculation, not just the final number.

Model + policy

What does P(s' | s,a) mean?

Bellman operations

Why sum outcomes but maximize actions?

Why terminal V = 0? Why $V_1 \neq V_2$? How do VI and PI differ?

OPTIONAL / DEEPER DISCUSSION

Why value iteration converges

Finite model, bounded rewards, $0 \le \gamma < 1$.

$$(TV)(s) = \max_a Q_V(s,a)$$

$$\|TV - TW\|_\infty \le \gamma \|V - W\|_\infty$$

$$\|V_k - V^*\|_\infty \le \gamma^k \|V_0 - V^*\|_\infty$$

A contraction has a unique fixed point: TV* = V*.

OPTIONAL / DEEPER DISCUSSION

Stopping and numerical error

A small update is evidence of a small residual.

$$\delta(V) = \|TV - V\|_\infty$$

$$\|V - V^*\|_\infty \le \delta(V) \,/\, (1-\gamma)$$

$$\Delta_k = \|V_{k+1} - V_k\|_\infty$$

$$\|V_{k+1} - V^*\|_\infty \le \gamma\Delta_k \,/\, (1-\gamma)$$

Label the returned vector correctly; the two bounds use different indices.

OPTIONAL / DEEPER DISCUSSION

A stochastic policy worked example

At start choose safe or risky with probability 0.5 each.

$$\pi(\text{safe} \mid \text{start}) = \pi(\text{risky} \mid \text{start}) = 0.5$$

$$V^{\pi}(\text{detour}) = 8$$

$$V^{\pi}(\text{start}) = 0.5[-1+0.9(8)] + 0.5[0.8(10)+0.2(-10)]$$

$$= 0.5(6.2) + 0.5(6) = 6.1$$

First average action choices; within risky, average successor outcomes.

OPTIONAL / DEEPER DISCUSSION

A cyclic MDP needs repeated backups

One state, one action: wait earns +1 and returns to itself.

$$V = 1 + 0.9V \quad \Rightarrow \quad 0.1V = 1 \quad \Rightarrow \quad V = 10$$

$$V_0 = 0, \;\; V_1 = 1, \;\; V_2 = 1.9, \;\; V_3 = 2.71$$

$$V_k = (1 - 0.9^k) \,/\, (1 - 0.9)$$

Unlike the delivery example, values approach the limit over many sweeps.

OPTIONAL / DEEPER DISCUSSION

When does the decision switch?

Compare the two expected returns algebraically.

$$Q_{\text{risky}} = (1-p)10 + p(-10) = 10 - 20p$$

$$Q_{\text{safe}} = -1 + 8\gamma$$

$$\text{At } \gamma=0.9: \quad 10-20p = 6.2 \quad \Rightarrow \quad p = 0.19$$

$$\text{At } p=0.2: \quad -1+8\gamma = 6 \quad \Rightarrow \quad \gamma = 0.875$$

At equality both actions maximize expected return.